Piecewise linear graphs

1 Consider the following piecewise relationship:

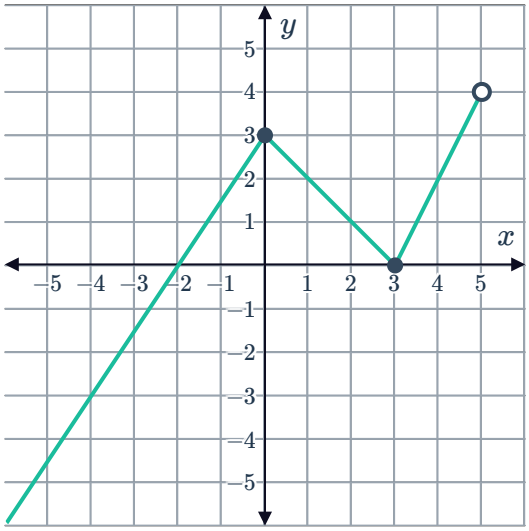
$$f(x) = \begin{cases} -4x - 6 & \text{when } x \leq a \\ x - 1 & \text{when } x > a \end{cases}$$

a If the graph of the piecewise relationship is connected, find the value of a.

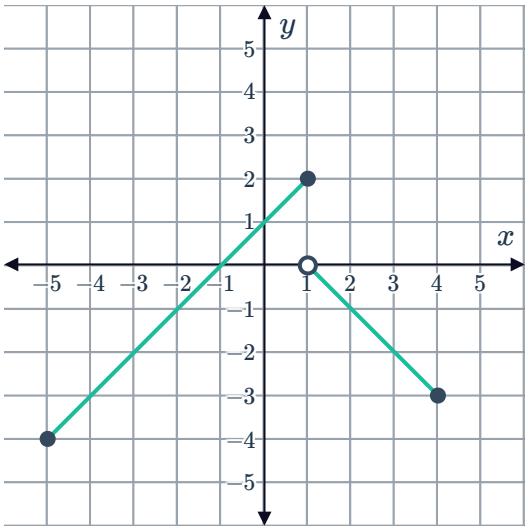
b Find the y-value of the piecewise relationship at $x = a$.

2 Write the piecewise function definition of the following graphs:

a



b



3 Sketch the following piecewise functions:

a $$f(x) = \begin{cases} -3, & x < 0 \\ x - 3, & x > 0 \end{cases}$$

b $$f(x) = \begin{cases} x + 2, & x < -4 \\ -2, & -4 \leq x < 4 \\ 2 - x, & x \geq 4 \end{cases}$$

c $$f(x) = \begin{cases} 5 - x, & x \leq -1 \\ 6, & -1 \leq x \leq 1 \\ x + 5, & x \geq 1 \end{cases}$$

d $$f(x) = \begin{cases} x + 4, & x < -3 \\ 1, & -3 < x < 2 \\ x - 1, & x \geq 2 \end{cases}$$

e $$f(x) = \begin{cases} 3, & x < 0 \\ x + 3, & x > 0 \end{cases}$$

f $$f(x) = \begin{cases} 4 - x, & x \leq -2 \\ 6, & -2 \leq x \leq 2 \\ x + 4, & x \geq 2 \end{cases}$$

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Piecewise Models

4 Sketch the graph that represents the following scenarios:

a A kitchen sink starts with $20\ \mathrm{L}$ of water. It empties at a rate of $4\ \mathrm{L}$ per minute and then the drain gets blocked. No water empties for 2 more minutes while it gets unblocked. Then the remaining water is drained out in 1 minute.

b A tank is being filled at a constant rate of $4\ \mathrm{L}$ per second for 4 minutes. The hose is then it is turned off for 2 minutes. Finally it is topped up again with water at a rate of $3\ \mathrm{L}$ per second for 1 minute.

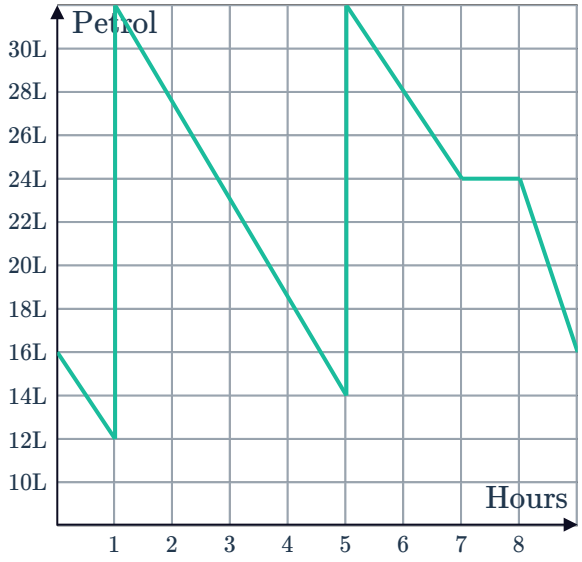
5 The line graph shows the amount of petrol (in litres) in a car's tank during a long drive: Given that the drive started at 8 am:

a How much petrol was initially in the tank?

b What happened at 9 am and 1 pm?

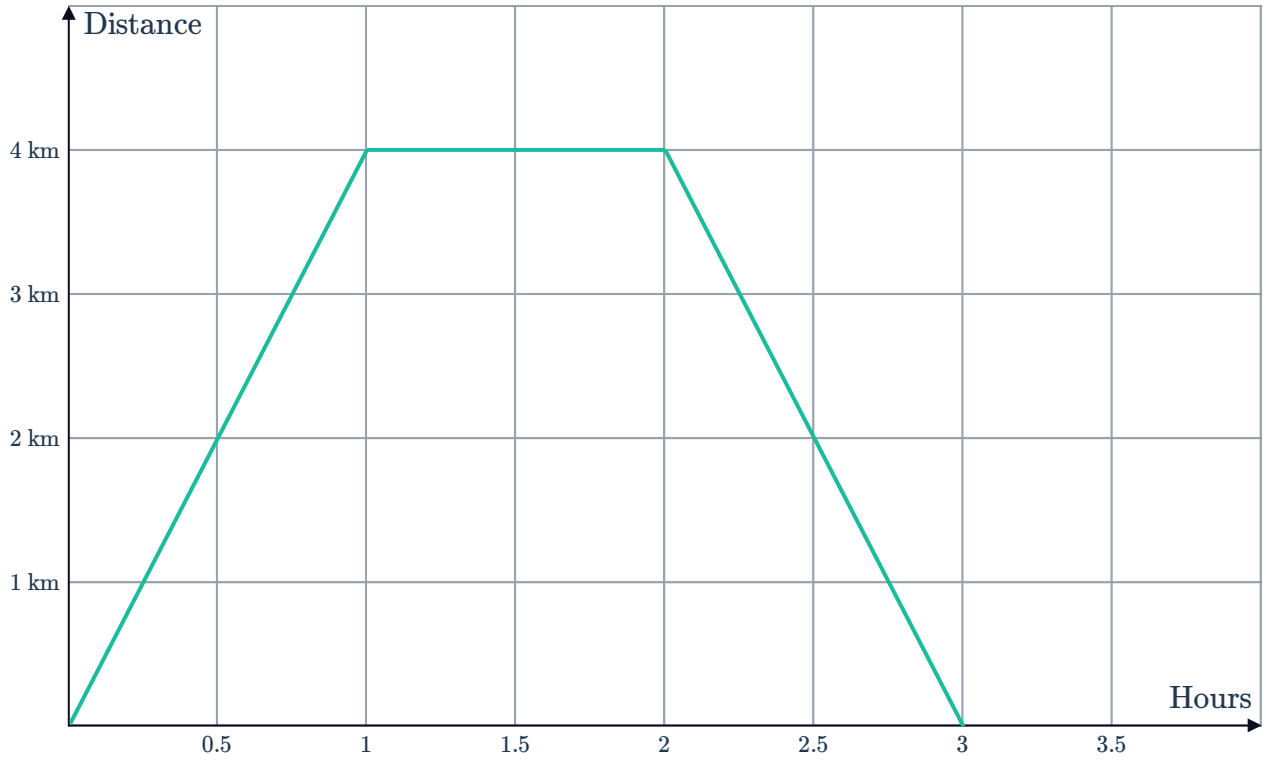
c How much petrol was used between 1 pm and 5 pm?

d To the nearest hour, when did the petrol in the tank first fall below 18 litres?



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6 The following line graph shows the distance Buzz was from his house as he travelled to visit his grandparents:



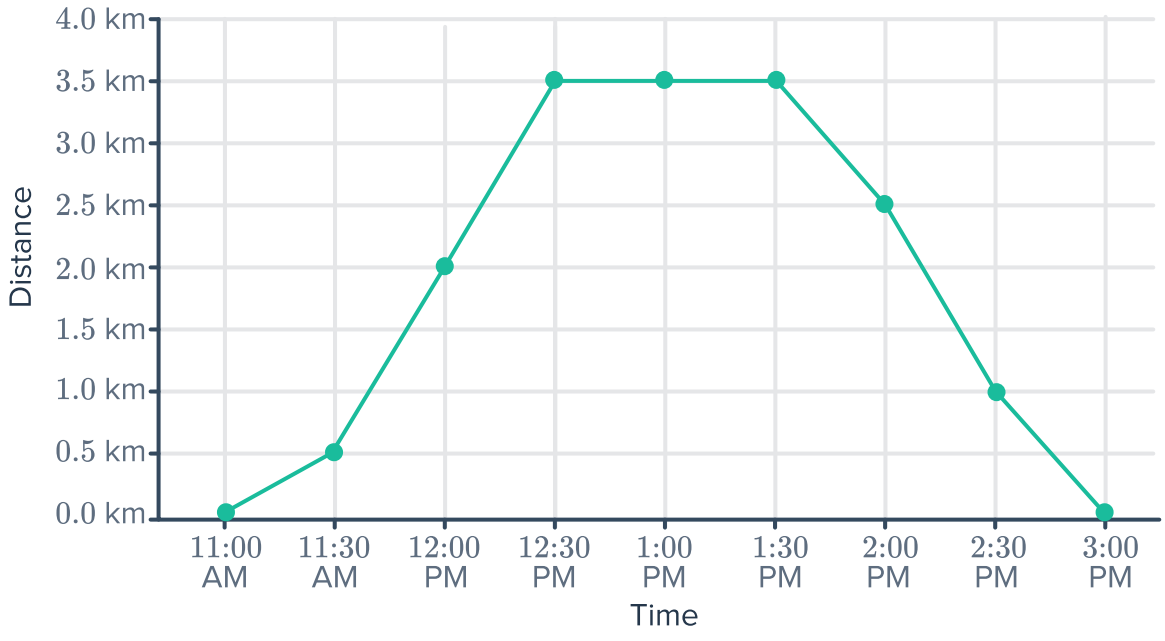
Given that Buzz left home at 10 am, find:

a How far Buzz was from his house at $10:30$ am.

b The time Buzz got back home.

c The furthest distance Buzz was from his house.

7 Danielle's distance from home throughout the day is recorded on the line graph below.



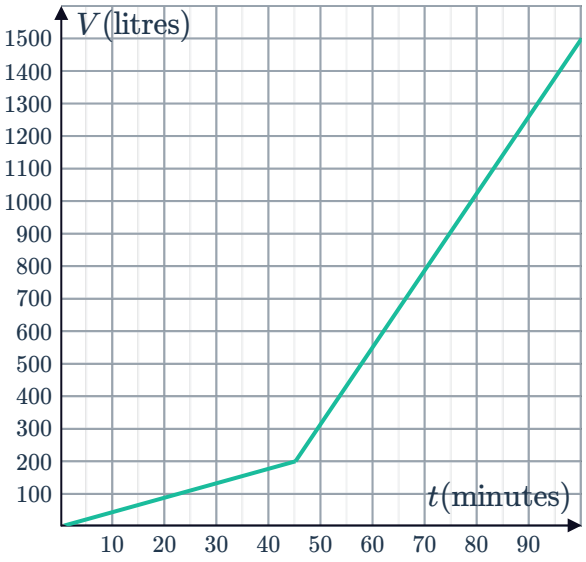
a How far was Danielle from her house at 2:00 pm?

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b What time did Danielle get back home?

c What was the furthest distance Danielle was from her house?

8 A children's pool is being filled with water. The volume of water V, in litres, after t minutes is given by the given piecewise graph. The pool has a maximum capacity of $1500\ \text{L}$.



a Estimate the volume of water, in litres, after 45 minutes have elapsed, to the nearest 10 litres.

b Determine the equation that describes the volume of water V after t minutes for the first 45 minutes.

c After 45 minutes, the rate at which the volume of water enters the pool is increased.
The piecewise relationship after $t = 45$ is given by $V = 26t - 970$.
Determine the time, t in minutes, when the pool is filled up.

9 A marathon runner initially runs at an average pace of $15\ \text{km/h}$. After 40 minutes of running at this speed, the runner slows down to a more comfortable pace of $12\ \text{km/h}$ for the remaining 160 minutes.
Let D be the distance covered by the marathon runner in kilometres and t be the time elapsed in minutes.

a State the piecewise relationship for D and t

b Find the distance covered in kilometres after each amount of time has elapsed:

i 20 minutes **ii** 40 minutes **iii** 60 minutes **iv** 200 minutes

c Graph the piecewise relationship.

10 Harry goes out for a run. He accelerates from rest up to a desired speed and maintains that speed for some time. Feeling exhausted, his velocity drops until he's back at rest.
The speed S in km/h after t seconds is given by the following piecewise relationship:

$$S = \begin{cases} 0.8t & \text{when } 0 \leq t < 15 \\ 12 & \text{when } 15 \leq t < 240 \\ 252 - t & \text{when } 240 \leq t \leq 252 \end{cases}$$

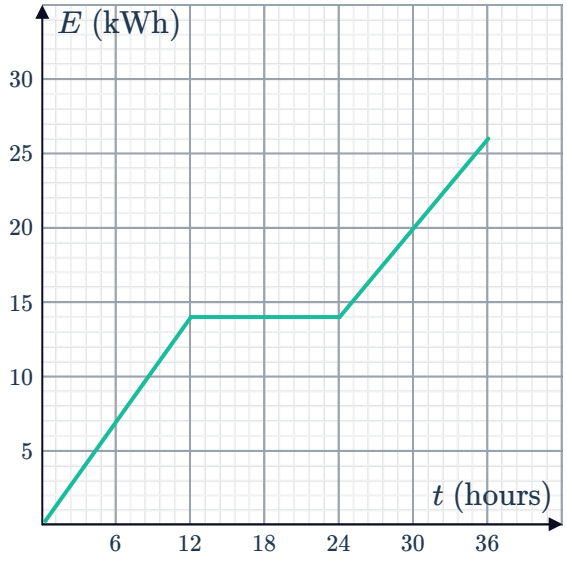
a Find the speed in km/h after the following times have elapsed:

i 10 seconds **ii** 50 seconds **iii** 150 seconds **iv** 250 seconds

b Graph the piecewise relationship.

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11 The amount of energy stored E in kWh in a set of solar batteries t hours after 6:00 am is given by the piecewise graph below. The storage is shown for a 36 hour period.



a Part of the piecewise graph is given by $E = \frac{7}{6}t, 0 \leq t \leq a$. Determine the value of a.

b After 12 hours have elapsed, the amount of energy stored is 14 kWh. Determine the equation for the part of the piecewise graph over the region $12 < t \leq 24$.

c The equation that describes the last piece of the piecewise graph is $E = t - 10$. State the time interval that the last piece is defined over.

12 Zlatko is competing in a swimming race being held in the ocean. From the shore, he swims 500 m out to a buoy in the ocean and then returns to the shore.
Below is the relation that describes the Zlatko's swim, where D is the distance from the shore, in metres, t minutes after the race begins. F is the time it took for Zlatko to finish the race.

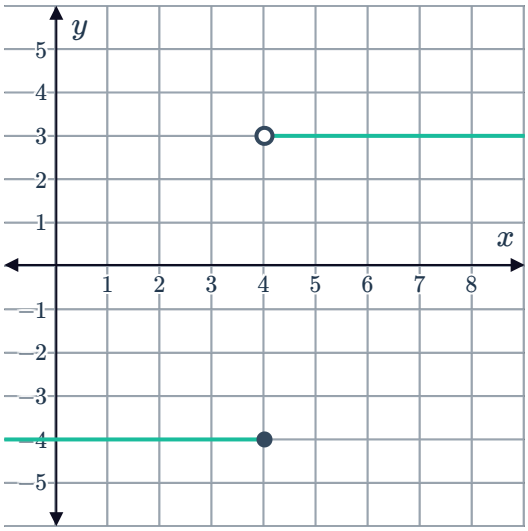
$$D(t) = \begin{cases} 50t, & 0 < t \leq 10 \\ -62t + 1120, & 10 < t \leq F \end{cases}$$

a Sketch the piecewise linear graph that shows Zlatko's swim.

b What distance, in metres, did Zlatko swim in the first 14 minutes of the race?

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15 Consider the following step graph:

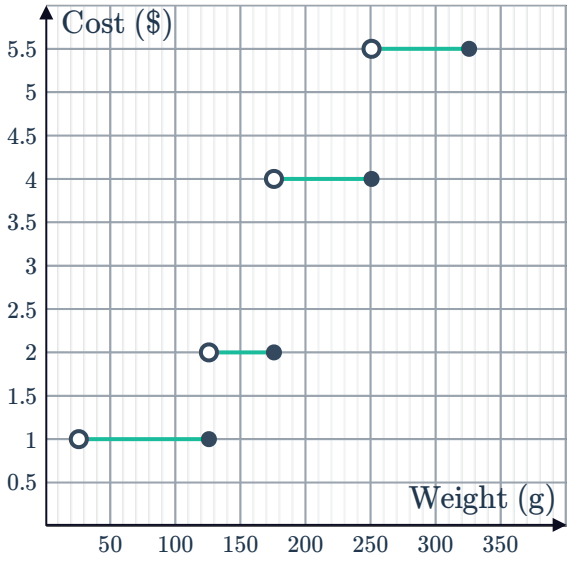


a State the equation of the line for $x > 4$.

b State the equation of the line for $x \leq 4$.

Step graph models

16 The graph shows the cost of sending parcels of various weight overseas:



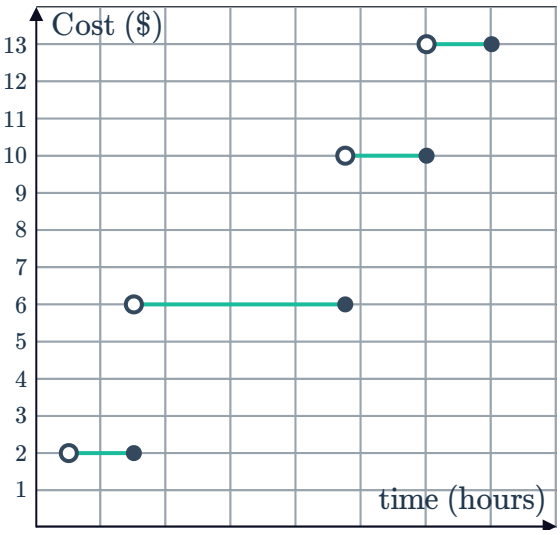
a Find the cost of sending a letter weighing:

i 100 g

ii 300 g

b Find the heaviest letter that can be sent for $2.

17 The graph shows the parking costs over different lengths of time:



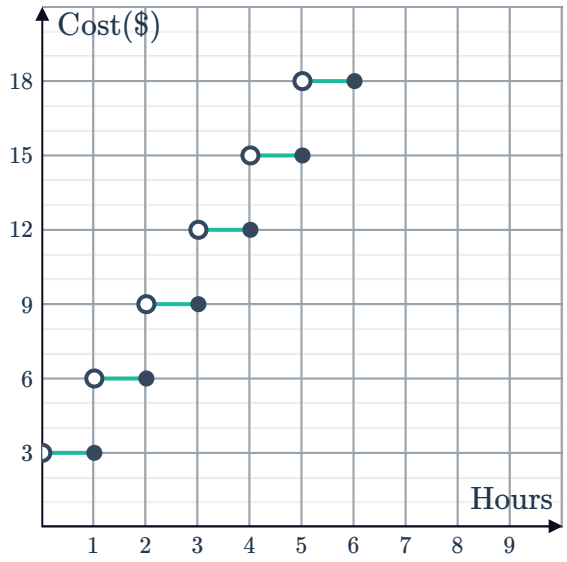
a How much would 5 hours of parking cost?

b Find the longest you can park your car for $6.

c The carpark offers a weekly pass for $44. If Aaron parks his car for 6 hours each day, five days a week, how much would he save each week with the weekly pass?

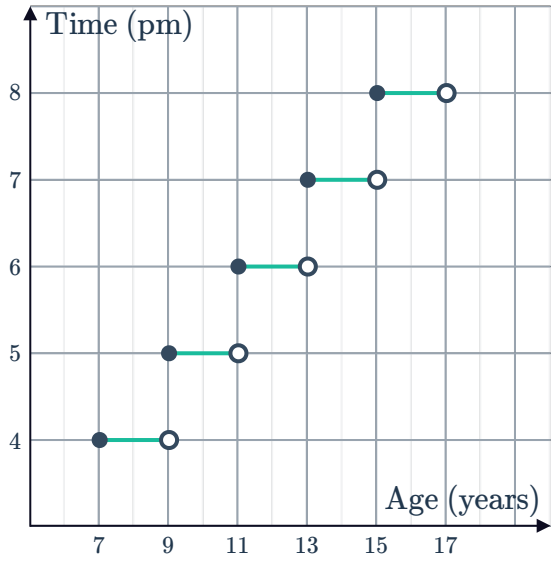
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18 The graph shows the amount (in dollars) an internet cafe charges its customers:



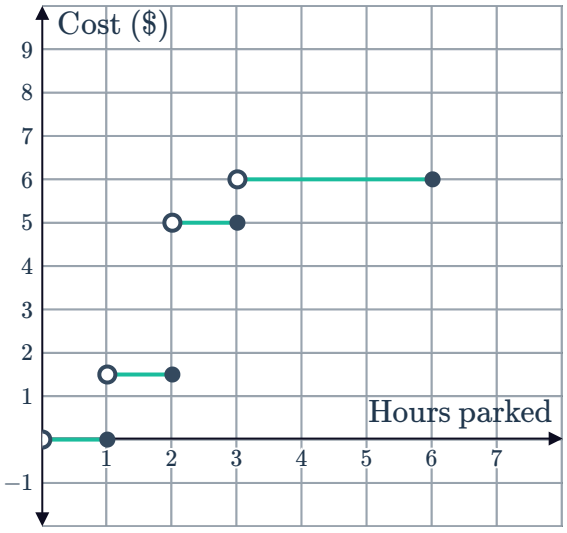
- **a** How much does Parvaneh have to pay if she uses their internet service for 2 hours and 40 minutes?
- **b** How much does Parvaneh have to pay if she uses their internet service for 4 hours?
- **c** Find the maximum number of hours that Parvaneh can use for $\$15$.

19 The graph shows the starting times of a shot put event for participants of different age groups:



- **a** What time will a participant who is aged 13 start their race?
- **b** Do any of the races start at half past the hour?

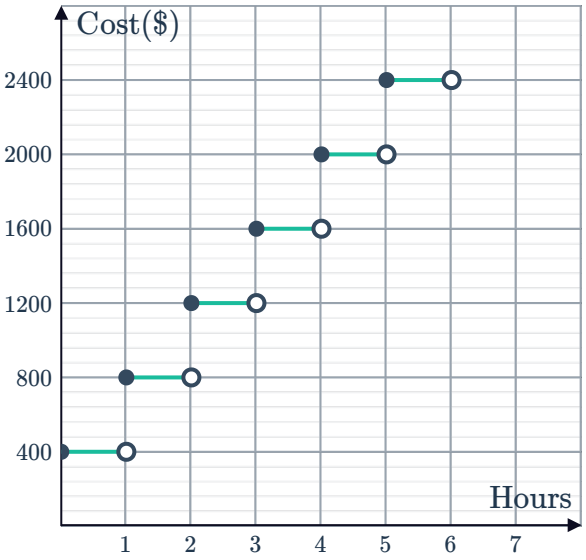
20 The graph shows the total cost of parking as a function of the number of hours parked:



- **a** How many free hours does this parking garage offer?
- **b** Peyvand goes to a movie and parks for 2.5 hours. How much will it cost her?
- **c** Tirdad works at the cinemas and parks for his entire 4.5-hour shift. How much will it cost him in parking?

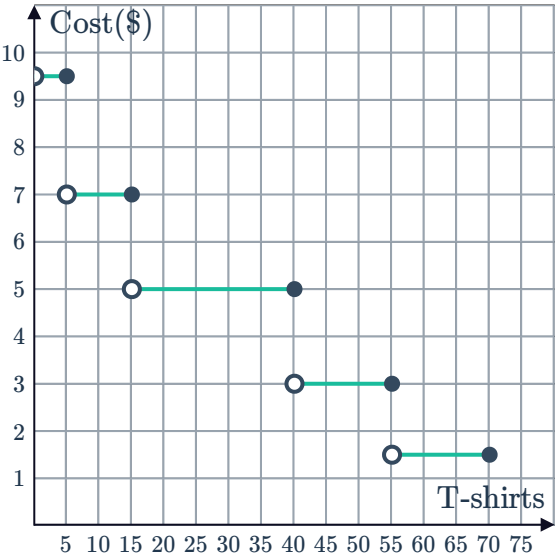
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21 The graph shows the amount a lawyer charges for consultations:



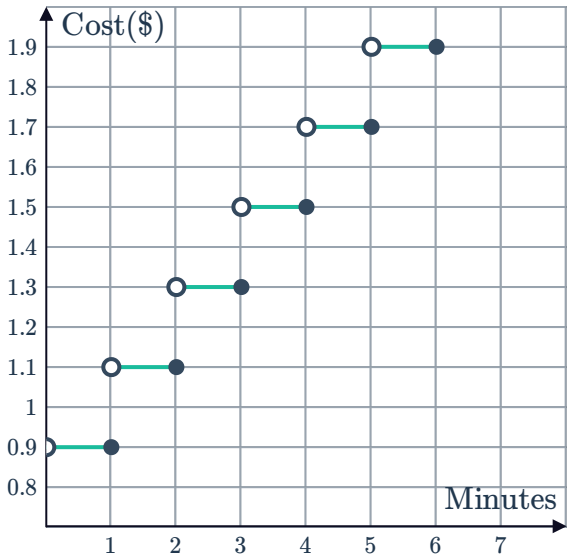
- **a** How much does the lawyer charge for a 5.5 hour consultation?
- **b** How much does the lawyer charge for a 4 hour consultation?
- **c** Find the shortest possible consultation that the lawyer will charge $\$800$ for.

22 The graph shows the cost per t-shirt when purchasing in bulk:



- **a** Determine the cost per t-shirt when 45 t-shirts are bought in bulk.
- **b** Determine the cost per t-shirt when 55 t-shirts are bought in bulk.
- **c** Find the most number of t-shirts that can be purchased at $\$7.00$ each.

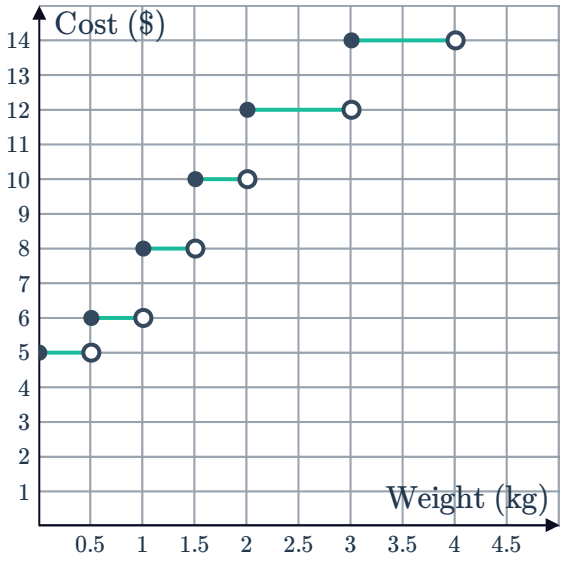
23 The graph shows the cost of a mobile phone call as a function of the length of the call:



- **a** How much does a call that lasts 4 minutes and 5 seconds cost?
- **b** How much does a 3-minute call cost?
- **c** Find the longest possible call that could be made for $\$1.50$.
- **d** Find the cost of each additional minute.
- **e** Find the initial connection cost.

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24 The step graph shows the cost for postage of packages based on their weight:



a Find how much it will cost to post a package that weighs:

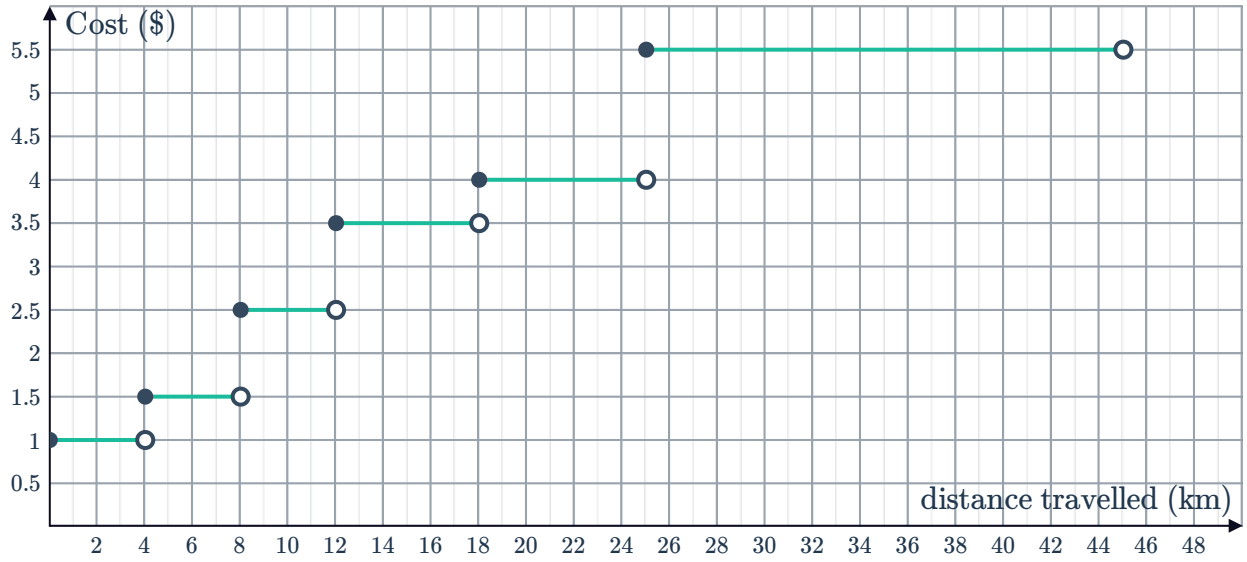
i 1.7 kg **ii** 2.5 kg **iii** 0.5 kg **iv** 3.2 kg

b If Neil has $11, what range of package weights that he can afford to send?

c Find the range of package weights can be sent for:

i $6 **ii** $7 **iii** $5.50 **iv** $14

25 The cost of a train ticket, based on distance travelled, is indicated by the step graph below:



a If Valentina needed to travel a distance of 11 km, how much would her train ticket cost?

b Luke has a doctor's appointment 19 km away. How much money will he need for a round-trip ticket (there and back again)?

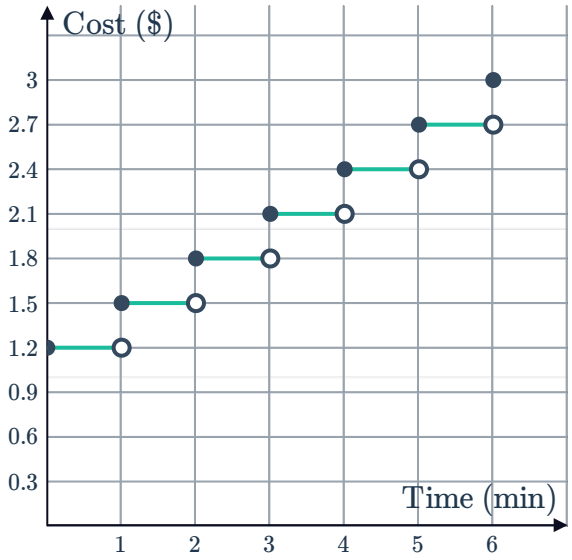
c Sally needs to travel 4 km to get to the grocery store. How much will a one way ticket cost?

d With $3, if Dave buys a round-trip ticket (there and back again), he can travel no further than what distance?

e Maria commutes approximately 40 km each direction every day to work. Calculate the total cost for one week's worth (5 days) of round-trip tickets.

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26 The step graph shows the charges for a home phone plan:



- **a** How much will a 3-minute phone call cost?
- **b** How much will 4 phone calls, each longer than 1 minute but less than 2 minutes, cost?
- **c** A phone line user made 12 calls that were just under 2 minutes long each, 13 calls just under 4 minutes long each, and 5 calls of less than 1 minute each. Calculate the total charges for the month.

27 Hiring helium tanks for balloons at a party costs $\$20$ up to and including the first hour, and $\$5$ per extra hour or any part thereof.
Sketch the graph that represents the total cost for any number of hours.

28 A telephone call is charged at $\$0.70$ for a call length of less than 1 minute, and $\$0.20$ extra per minute after that.
Graph the cost of calls up to 5 minutes in length, where the vertical axis is in dollars.

29 The cost of a train fare is calculated at $\$1.30$ for the first zone and then $20c$ per additional zone after that.
Graph the fares for trips up to five zones, where the vertical axis is in dollars.

30 At an indoor ski facility, the temperature is set to -5°C at 2 pm. At 3 pm, the temperature is immediately brought down to -12°C and left for 3 hours before immediately taking it down again to -18°C, where it stays for the rest of the day's operation. The facility operates until 10 pm.

- **a** Write a stepwise function that models the indoor temperature, y, at a certain time of the day, x hours after midday.
- **b** Create a graph of the step function relating time of day and temperature inside the ski facility.
- **c** Lakota entered the ski facility at $3:30$ pm. What was the temperature inside the facility at this time?
- **d** Xavier wants to wait till the indoor temperature is -7°C or lower. When is the earliest he can enter the facility?

31 A plumber charges a call-out fee of $\$20$ to come out to the site and on top of that, charges $\$10$ per 15 minute block on site or part thereof.
Create a step graph that matches this scenario, where the vertical axis is in dollars.

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32 Create a step graph for the charge for sending parcels as shown in the table:

Express post satchels	Maximum weight (g)	Charge per item
Small (220×355)	500	\$2.50
Medium (310×405)	3000	\$4.25
Large (435×510)	5000	\$7.75

33 The government is looking at alternative taxation schemes and one proposal is to pay a fixed amount of tax depending on which income bracket you fall into.
The function below models the tax payable, y, based on an income of x:

$$y = \begin{cases} \$1200, & \$0 < x \leq \$15\,000 \\ \$4900, & \$15\,000 < x \leq \$35\,000 \\ \$12\,350, & \$35\,000 < x \leq \$65\,000 \\ \$43\,320, & \$65\,000 < x \leq \$114\,000 \\ \$54\,720, & x > \$114\,000 \end{cases}$$

a Under this proposal, how much tax will you pay on an income of \$14 000?

b What is the difference in income between the highest income earner and lowest income earner who are both paying \$43 320 in tax? Assume you are taxed on whole dollars you earn.

c Estelle is earning an income of \$63 800. She is set to get a pay rise at the end of the year. Find the maximum pay rise she can get before falling into the next highest tax bracket.

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